

Cameron Morgan | PhD Candidate

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Research interests

Galaxy evolution, pre-processing in galaxy groups, environmental quenching of star formation, galaxy star formation histories, evolution of galaxy groups and clusters, galaxy morphology, next-generation space telescopes

Academic Positions

Postdoctoral Researcher

St. Mary's University

Halifax, Canada

September 2026–

Visiting Postdoctoral Researcher

NRC Herzberg Astronomy and Astrophysics Research Centre

Victoria, Canada

September 2026–

Education

Doctor of Philosophy in Physics and Astronomy

University of Waterloo

Supervisor: Dr. Michael Balogh

PhD thesis: "Galaxy evolution across the Virgo cluster environment"

Waterloo, Canada

September 2022–August 2026 (expected)

Master of Science in Physics and Astronomy

University of Waterloo

Supervisor: Dr. Michael Balogh

MSc thesis: "Ultraviolet Imaging and Morphology of Galaxies in the Coma Cluster"

Waterloo, Canada

September 2020–August 2022

Honours Bachelor of Science in Physics and Astronomy (co-op)

University of Waterloo

Dean's Honour List

Waterloo, Canada

September 2015–June 2020

Publications

As Lead Author

2. Morgan, C.R., Sazonova, E., Roberts, I.D., Balogh, M.L. et al., 2025, *Deep Galaxy Stellar Mass Functions As a Function of Star Formation Rate in the Virgo Cluster Environment*. The Astrophysical Journal, 987, 166.

1. Morgan, C.R., Balogh, M.L., Boselli, A., Fossati, M. et al., 2024, *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XVI. the ubiquity of truncated star-forming disks across cluster substructures*. Astronomy & Astrophysics, 691, 20.

As Co-author

7. Sazonova, E., Morgan, C.R., Balogh, M., 2026, *The Hubble sequence in JWST CEERS from unbiased galaxy morphologies*. Submitted to The Open Journal of Astrophysics.

6. Sazonova, E., Morgan, C.R., Balogh, M. et al., 2025, *statmorph-lsst: quantifying and correcting morphological biases in galaxy surveys*. The Open Journal of Astrophysics, 9.

5. Lawlor-Forsyth, C., Balogh, M.L., Sazonova, E., **Morgan, C.R.** et al., 2025, *Identifying and distinguishing quenching galaxies with spatially-resolved star formation in TNG50*. The Astrophysical Journal, 1000, 61.
4. Côté, P., Woods, T., Hutchings, J.B. et al. including **Morgan, C.R.**, 2025, *The CASTOR mission*. Journal of Astronomical Telescopes, instruments, and Systems, 11, 4.
3. Martocchia, S., Boselli, A., Maraston, C. et al. including **Morgan, C.R.**, 2025, *A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE): XVIII. Reconstructing the star formation history of early-type galaxies through the combination of their UV and H α emission*. Astronomy & Astrophysics, 696, 79.
2. Sazonova, E., **Morgan, C.R.**, Balogh, M. et al., 2024, *RMS Asymmetry: a robust metric of galaxy shapes in images with varied depth and resolution*. Open Journal of Astrophysics, 7, 77.
1. Cheng, I., Woods, T.E., Côté, P. et al. including **Morgan, C.R.**, 2024, *FORECASTOR – I. Finding Optical Requirements and Exposure times for the Cosmological Advanced Survey Telescope for Optical and UV Research mission*. The Astronomical Journal, 167, 178.

Awards and Scholarships

Competitive Scholarships.....

Ontario Graduate Scholarships (2024-2026)

Awarded: 2024–2025, 2025–2026

Total grant amount: \$30,000

President's Graduate Scholarships, University of Waterloo (2024-2026)

Awarded: 2024–2025, 2025–2026

Total grant amount: \$10,000

Other Awards and Honours.....

Canadian Student Participation in Space Conferences and Training Events, Canadian Space Agency (\$1500, 2026)

Waterloo Centre for Astrophysics Best Student Paper Prize Winner, University of Waterloo (\$500, 2025)

Waterloo Centre for Astrophysics Best Student Paper Prize Honourable Mention, University of Waterloo (2025)

Graduate Student Research Dissemination Award, University of Waterloo (\$500 each in 2024, 2025)

Canadian Astronomical Society Graduate Student Committee "Best Talk" (2025)

Waterloo Centre for Astrophysics Outreach Award (2024, 2025)

Technical Reports

1. Patrick Côté, Tyrone Woods, John Hutchings and 135 others including **Cameron Morgan**, 2023, *Phase 0 Science Development Report for CASTOR*. Technical report submitted to the Canadian Space Agency.

Accepted Observing Proposals

As Principal Investigator.....

2. **Morgan, C.R.**, Boselli, A., Balogh, M. L., Fossati, M. et al., CFHT MegaCam, Semester 2025A, 31 hr, *Pre-processing in the W cloud: environmentally-driven evolution of galaxies prior to infall into the Virgo cluster*

1. **Morgan, C.R.**, Boselli, A., Balogh, M. L., Fossati, M., Vollmer, B., Epinat, B. and Roediger, J., CFHT MegaCam, Semester 2024A, 16.8 hr, *Pre-processing of galaxies in the W cloud: environmentally-driven evolution of galaxies prior to infall into the Virgo cluster*

As co-Investigator.....

2. Boselli, A., et al. including **Morgan, C.R.**, CFHT MegaCam, Semester 2025A, 31 hr, *Pre-processing in the W cloud, an infalling group in the Virgo cluster*
1. Boselli, A., et al. including **Morgan, C.R.**, CFHT MegaCam, Semester 2024A, 11 hr, *Pre-processing in the W cloud, and infalling group in the Virgo cluster*

Student Supervision

Isaac Cheng (undergraduate), University of Waterloo (primary supervisor)

Invited Talks and Seminars

3. *Quenching in and around the Virgo Cluster: a spatially resolved analysis*, 2026, University of Kansas Virgo Meeting. Lawrence, KS (attended remotely).
2. *Probing galaxy quenching in the Virgo cluster and beyond*, 2025, University of British Columbia Cosmology Journal Club. Vancouver, BC.
1. *Deep Galaxy Stellar Mass Functions As a Function of Star Formation Rate in the Virgo Cluster Environment*, 2025, York University/University of Toronto Galaxies Journal Club. Toronto, ON (attended remotely).

Contributed Talks and Posters

9. *Local star formation morphologies of Virgo cluster galaxies*, 2026, Canadian Astronomical Society Annual General Meeting (CASCA). Montréal, Canada. Contributed talk.
8. *Decoding environmentally-driven quenching of local star forming galaxies in the Virgo cluster*, 2026, American Astronomical Society Winter Meeting 247. Phoenix, USA. Contributed dissertation talk.
7. *The many lives of Virgo cluster galaxies*, 2025, Galaxy Evolution and Environment in Southern and Eastern Ontario (GEESE-ON). Waterloo, Canada. Contributed talk.
6. *Decoding quenching in the Virgo cluster with spatially resolved star formation*, 2025, Canadian Astronomical Society Annual General Meeting (CASCA). Halifax, Canada. Contributed talk - graduate student committee "best talk" winner.
5. *Spatially resolved star formation in Virgo cluster galaxies: decoding environmental quenching*, 2024, Star formation across the scales: star clusters to galactic disks. Hamilton, Canada. Contributed talk.
4. *Decoding quenching in the Virgo cluster and infalling groups with spatially resolved star formation*, 2024, Canadian Astronomical Society Annual General Meeting (CASCA). Toronto, Canada. Poster.
3. *Decoding quenching in the Virgo cluster and infalling groups with spatially resolved star formation*, 2024, GASPIA2024: The physical processes shaping the stellar and gaseous histories of galaxies. Pisa, Italy. Contributed talk.
2. *H α morphology in the Virgo Cluster with VESTIGE: the ubiquity of truncated star forming disks across cluster substructures*, 2023, Canadian Astronomical Society Annual General Meeting (CASCA). Penticton, Canada. Poster.
1. *Ultraviolet sizes and morphology of galaxies in the Coma cluster*, 2022, Canadian Astronomical Society Annual General Meeting (CASCA). Online. Poster.

Service

Co-chair, LOC/SOC, Galaxy Evolution and Environment in Southern and Eastern Ontario (GEESE-ON) 2025

Invited referee for *The Astrophysical Journal*

2025 – present

Invited referee for *Astronomy & Astrophysics*

2025 – present

Waterloo Centre for Astrophysics Graduate Student Representative

University of Waterloo

2023 – 2024

As the graduate student representative for the Waterloo Centre for Astrophysics, I represent all graduate students at monthly meetings where I voice ideas and concerns from the grad student body. I initiate discussions with the leaders of the centre to brainstorm ideas of how we can continually improve the Centre for graduate students and ultimately all members.

Physics Graduate Student Association / Waterloo Centre for Astrophysics Liaison

University of Waterloo

2023 – present

I am a member of the PhysGSA Executive Team as the liaison for the WCA. As part of this team, we manage funds to host social events for graduate students in Physics and maintain a graduate lounge space, solicit feedback on graduate student life and meet with Department leaders to bring about positive change to the grad program.

Outreach

Gustav Bakos Observatory Tour Lead and Presenter

University of Waterloo

2022 – present

I host talks and tours of the Gustav Bakos Observatory for the public as well as for private community groups. These tours are well-attended by the Waterloo community and allow me to share my passion for astronomy with those from a variety of backgrounds and experiences. In addition, we run activities and workshops for broader outreach events at the University, aimed at encouraging younger students to consider pursuing careers in STEM.

Outreach Volunteer and Coordinator, Waterloo Centre for Astrophysics

University of Waterloo

2023 – present

I am part of a team of grad student and postdoctoral volunteers at the Waterloo Centre for Astrophysics who host a variety of outreach and community engagement events including hosting mobile planetarium shows, Astronomy on Tap, talks at the Kitchener Public Library, backyard observation sessions during meteor shows and more. Currently, I am in charge of managing our monthly seminar series at the Kitchener Public Library.

Public Talks

5. *Canada's role in the future of space astronomy*, 2025, Astronomy on Tap: Kitchener-Waterloo. Kitchener, Canada.
4. *Love thy (galactic) neighbour: a journey through the local group*, 2025, Astronomy on Tap: Kitchener-Waterloo. Kitchener, Canada.
3. *Galactic mortality: the lives of galaxies and how they die*, 2024, Royal Astronomical Society of Canada - Kitchener-Waterloo Chapter. Waterloo, Canada.
2. *CASTOR: Canada's flagship space telescope*, 2024, KPL-WCA Astronomy Series. Kitchener, Canada.
1. *Galaxies Across the Spectrum: Observing Cosmic Habitats*, 2024, Astronomy on Tap: Kitchener-Waterloo. Kitchener, Canada.

Media Appearances

3. **Cam Morgan wins CASCA 2025 GSC award for best talk** <https://uwaterloo.ca/astrophysics-centre/news/cam-morgan-wins-casca-2025-gsc-award-best-talk>
2. **Waterloo Centre for Astrophysics Research Shorts** <https://www.youtube.com/watch?v=TFMMKM6fapE&t=1s>
1. **Humans of Waterloo: The Grad Students** <https://publications.uwaterloo.ca/humans-of-waterloo-grad/cameron-morgan.html>

Memberships

American Astronomical Society (AAS)	2025 – present
Canadian Astronomical Society (CASCAS)	2020 – present
Waterloo Centre for Astrophysics (WCA)	2020 – 2026

Collaborations

Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE)	2026 – present
Virgo Filaments Survey (VFS)	2026 – present
Gemini Infrared Multi-Object Spectrograph (GIRMOS), GTO science team	2025 – present
The Cosmological Advanced Survey Telescope for Optical and ultraviolet Research (CASTOR), Phase 0	
Galaxies Science Working Group member	2022 – present
A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE)	2022 – present
Next Generation Virgo cluster Survey (NGVS)	2022 – present

Research Experience

Graduate Research (PhD) **University of Waterloo**
Supervisor: Dr. Michael Balogh 2022-2026

A Virgo Environmental Survey of Ionised Gas Emission (VESTIGE: P.I. Alessandro Boselli). My current project involves using spatially resolved H α emission and other multi-band photometry of the Virgo cluster to make measurements of the morphology and resolved star formation histories of galaxies. These measurements allow us to make inferences into the various environmental quenching mechanisms shutting down star formation in these cluster galaxies. This has led to a paper on disk truncation published in A&A, and a paper on stellar mass functions published in ApJ, with ongoing work on resolved star formation histories.

Graduate Research (MSc) **University of Waterloo**
Supervisor: Dr. Michael Balogh 2020-2022

Using high-resolution ultraviolet imaging of the Coma cluster obtained with the Ultraviolet Imaging Telescope (UVIT, P.Is. Balogh and Côté), we examined the presence of truncated star-forming disks of galaxies being altered by the cluster environment. The disk sizes of galaxies were measured using a combination of Sérsic profile fitting techniques (GALFIT, STATMORPH) as well as flux growth curve extrapolation. We find that star-forming galaxies in the Coma cluster has smaller UV disk sizes when matched to field galaxies of similar stellar mass and colour. We also find a high fraction of red galaxies in Coma, indicating that quenching processes in the cluster are shutting down star formation before affecting the galaxy morphology.

Undergraduate Research **University of Waterloo**
Supervisor: Dr. Michael Balogh 2020

First look at UV imaging of the Coma cluster with the Ultraviolet Imaging Telescope (UVIT, P.Is. Balogh and Côté). Source extraction was performed using customized techniques specific to ultraviolet data.

Research Assistant **Environment and Climate Change Canada**
Supervisor: Dr. Zen Mariani 2018

As a co-op student, I worked as part of a team remotely monitoring and testing novel weather sensing instruments employed in Canada's arctic region (Iqaluit, Nunavut and Whitehorse, Yukon Territory). My role involved receiving data from these instruments and writing code to produce data products to monitor the output. I also authored a report to senior management on the first light of a far-infrared radiometer deployed in Iqaluit.

Teaching experience

Graduate Teaching Assistant, University of Waterloo

2020–2026

PHYS 275, 370L, 474

Duties include grading assignments, exams and reports from students (all courses). In addition, I hold office hours where students seek advice and help on assignments. For PHYS 370L, I was responsible for hosting observing sessions for students at the Gustav Bakos Observatory on campus, where students had the chance to take astronomical images of various objects as data collection for experiments. As such, I was also responsible for maintaining and ensuring safe use of equipment including a telescope, CCD camera and dome. I have also given occasional lectures to fill in for absent instructors.

Math Tutor and Learning Centre Associate, Humber and Centennial College

2017–2018

I oversaw the operations of a math tutoring centre for college students enrolled in a variety of math courses. The tutoring centre was free for students to drop in where tutors including myself provided assistance on homework and test prep and ran impromptu tutorials.